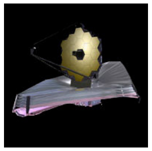

3153 - Why do some 50 Myr old stars still accrete?

Cycle: 2, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1		MIRI Medium Resolution Spectroscopy	(1) WISEA-J080822.18-644357.3
	2		MIRI Medium Resolution Spectroscopy	(2) 2MASS-J05010082-4337102
	3		MIRI Medium Resolution Spectroscopy	(3) 2MASS-J02265658-5327032
	10		MIRI Medium Resolution Spectroscopy	(10) Group J0446AB
	11		MIRI Medium Resolution Spectroscopy	(11) Group J0949AB
	8		MIRI Medium Resolution Spectroscopy	(8) 2MASS-J06320799-6810419
	9		MIRI Medium Resolution Spectroscopy	(9) 2MASS-J15460752-6258042

ABSTRACT

The outcome of planet formation is largely determined by how the disk dust and gas evolve. Recent studies have identified a new class of disks that surround late M type stars (>M4.5) at ages of 40-60 Myr and with spectroscopic evidence of active accretion. The strong infrared excess and broad H alpha emission lines are hallmarks of gas-rich primordial disks. Given the typical disk lifetime of 3-5 Myr, surviving for ~10 times longer is very surprising. If these are indeed primordial disks, their survival would revise our understanding of disk dissipation and planet formation around M dwarfs. On the other hand, their infrared excess levels are also consistent with a type of unusually dust-rich debris disks that are believed to have experienced a recent giant collision, along with more compatible ages. We propose to use MIRI/MRS to study the gas and dust content in these peculiar disks and to determine their evolutionary status. The detection of abundant C-bearing molecules with a high C/O ratio, characteristic of young M dwarf disk chemistry, would suggest a long-lived gas-rich disk, while the presence of a large amount of silica dust would point to a recent high-energy collision. Both scenarios will have important implications for planet formation, as the former provides clues for the build-up process of compact systems like TRAPPIST-1, while the latter dives into the active stage of terrestrial planet assembly.

OBSERVING DESCRIPTION

This program aims at obtaining MIRI/MRS spectra for 9 old accreting M dwarf disks, using all three gratings to cover the full wavelength range from 5 to 28 micron. We are particularly interested in the 10 and 20 micron dust solid-state features and molecular gas emission lines at around 14 micron. Exposure times have been optimized to detect the fainter line emission. We use FAST readout for all but one faint sources. We include target acquistition for all sources and choose the standard 4 point dither as the emission is expected to be unresolved.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	WISEA-J080822.18-644357.3	RA: 08 08 22.1821 (122.0924254d) Dec: -64 43 57.26 (-64.73257d) Equinox: J2000	Proper Motion RA: -0.0018464208314892905 sec of time/yr Proper Motion Dec: 0.025463 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[M dwarfs, Young stellar objects] Extended=NO				
(2)	2MASS-J05010082-4337102	RA: 05 01 0.8931 (75.2537212d) Dec: -43 37 9.94 (-43.61943d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[M dwarfs, Young stellar objects] Extended=NO				
(3)	2MASS-J02265658-5327032	RA: 02 26 56.7581 (36.7364921d) Dec: -53 27 3.46 (-53.45096d) Equinox: J2000	Proper Motion RA: 0.010348542411799187 sec of time/yr Proper Motion Dec: -0.01878200000646757 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Brown dwarfs] Extended=NO				
(4)	2MASS-J04463413-2627559-A	RA: 04 46 34.1066 (71.6421108d) Dec: -26 27 56.84 (-26.46579d) Equinox: J2000	Proper Motion RA: 33.525 mas/yr Proper Motion Dec: -3.475 mas/yr Epoch of Position: 2016.0	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. coordinates and proper motions adopt from Gaia DR3 Category=Star Description=[M dwarfs] Extended=NO				
(5)	2MASS-J04463413-2627559-B	RA: 04 46 34.2506 (71.6427108d) Dec: -26 27 55.57 (-26.46544d) Equinox: J2000	Proper Motion RA: 33.225 mas/yr Proper Motion Dec: -5.605 mas/yr Epoch of Position: 2016.0	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. coordinates and proper motions adopt from Gaia DR3 Category=Star Description=[M dwarfs, Young stellar objects] Extended=NO				
(6)	2MASS-J09490073-7138034-A	RA: 09 49 0.7488 (147.2531200d) Dec: -71 38 2.93 (-71.63415d) Equinox: J2000	Proper Motion RA: -36.117 mas/yr Proper Motion Dec: 27.85 mas/yr Epoch of Position: 2016.0	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. coordinates and proper motions adopt from Gaia DR3 Category=Star Description=[M dwarfs, Young stellar objects] Extended=NO				

(7)	2MASS-J09490073-7138034-B	RA: 09 49 0.4371 (147.2518213d) Dec: -71 38 3.15 (-71.63421d) Equinox: J2000	Proper Motion RA: -39.1017 mas/yr Proper Motion Dec: 23.5828 mas/yr Epoch of Position: 2016.0
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. coordinates and proper motions adopt from Gaia DR3 Category=Star Description=[M dwarfs, Young stellar objects] Extended=NO			
(8)	2MASS-J06320799-6810419	RA: 06 32 8.0052 (98.0333550d) Dec: -68 10 41.46 (-68.17818d) Equinox: J2000	Proper Motion RA: 0.0016768833489793644 sec of time/yr Proper Motion Dec: 0.033936 arcsec/yr Epoch of Position: 2015.5
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[M dwarfs, Young stellar objects] Extended=NO			
(9)	2MASS-J15460752-6258042	RA: 15 46 7.4264 (236.5309433d) Dec: -62 58 5.19 (-62.96811d) Equinox: J2000	Proper Motion RA: -0.006274628706688199 sec of time/yr Proper Motion Dec: -0.061790999939148605 arcsec/yr Epoch of Position: 2015.5
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[M dwarfs, Young stellar objects] Extended=NO			
(10)	Group J0446AB		
Comments: Target Selection=[4 2MASS-J04463413-2627559-A, 5 2MASS-J04463413-2627559-B]			
(11)	Group J0949AB		
Comments: Target Selection=[6 2MASS-J09490073-7138034-A, 7 2MASS-J09490073-7138034-B]			

Observation

Proposal 3153, Observation 1

Wed May 10 23:18:08 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	WISEA-J080822.18-644357.3	RA: 08 08 22.1821 (122.0924254d) Dec: -64 43 57.26 (-64.73257d) Equinox: J2000	Proper Motion RA: -0.0018464208314892905 sec of time/yr Proper Motion Dec: 0.025463 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[M dwarfs, Young stellar objects]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FAST	4	1	1	11.1	141558.14

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
1	SHORT(A)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	
1	SHORT(A)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	
2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	
2	MEDIUM(B)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	
3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008	
3	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	
3	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	

Observation

Proposal 3153, Observation 2

Wed May 10 23:18:08 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	2MASS-J05010082-4337102	RA: 05 01 0.8931 (75.2537212d) Dec: -43 37 9.94 (-43.61943d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[M dwarfs, Young stellar objects]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FAST	4	1	1	11.1	141558.14

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
1	SHORT(A)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	
1	SHORT(A)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	
2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	
2	MEDIUM(B)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	
3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008	
3	LONG(C)	MRSLONG		FASTR1	50	2	1	Dither 1	4	8	1121.116	
3	LONG(C)	MRSSHORT		FASTR1	50	2	1	Dither 1	4	8	1121.116	

Observation

Proposal 3153, Observation 3

Wed May 10 23:18:08 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	2MASS-J02265658-5327032	RA: 02 26 56.7581 (36.7364921d) Dec: -53 27 3.46 (-53.45096d) Equinox: J2000	Proper Motion RA: 0.010348542411799187 sec of time/yr Proper Motion Dec: -0.01878200000646757 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Brown dwarfs]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FAST	4	1	1	11.1	141558.15

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
1	SHORT(A)	MRSLONG		SLOWR1	20	2	1	Dither 1	4	8	3917.947	
1	SHORT(A)	MRSSHORT		SLOWR1	20	2	1	Dither 1	4	8	3917.947	
2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008	
2	MEDIUM(B)	MRSLONG		SLOWR1	20	2	1	Dither 1	4	8	3917.947	
2	MEDIUM(B)	MRSSHORT		SLOWR1	20	2	1	Dither 1	4	8	3917.947	
3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008	
3	LONG(C)	MRSLONG		SLOWR1	20	2	1	Dither 1	4	8	3917.947	
3	LONG(C)	MRSSHORT		SLOWR1	20	2	1	Dither 1	4	8	3917.947	

Observation

Proposal 3153, Observation 10

Wed May 10 23:18:08 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Comments: The aperture PA range is set for the dither offset direction, to avoid moving along the binary position angle

Diagnostics

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	Group J0446AB			

Comments:
Target Selection=[4 2MASS-J04463413-2627559-A, 5 2MASS-J04463413-2627559-B]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FAST	4	1	1	11.1	141558.16

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F770W	FASTR1	25	1	1	Dither 1	4	4	277.504	
1	SHORT(A)	MRSLONG		FASTR1	60	2	1	Dither 1	4	8	1343.119	
1	SHORT(A)	MRSSHORT		FASTR1	60	2	1	Dither 1	4	8	1343.119	
2		IMAGER	F1000W	FASTR1	25	1	1	Dither 1	4	4	277.504	
2	MEDIUM(B)	MRSLONG		FASTR1	60	2	1	Dither 1	4	8	1343.119	
2	MEDIUM(B)	MRSSHORT		FASTR1	60	2	1	Dither 1	4	8	1343.119	
3		IMAGER	F1130W	FASTR1	25	1	1	Dither 1	4	4	277.504	
3	LONG(C)	MRSLONG		FASTR1	60	2	1	Dither 1	4	8	1343.119	
3	LONG(C)	MRSSHORT		FASTR1	60	2	1	Dither 1	4	8	1343.119	

Special Requirements	
	Aperture PA Range 20 to 60 Degrees (V3 20.0 to 60.0)

Observation

Proposal 3153, Observation 11

Wed May 10 23:18:08 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Comments: The aperture PA range is set for the dither offset direction, to avoid moving along the binary position angle

Diagnostics

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(11)	Group J0949AB			

Comments:
Target Selection=[6 2MASS-J09490073-7138034-A, 7 2MASS-J09490073-7138034-B]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FAST	4	1	1	11.1	141558.17

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F770W	FASTR1	50	1	1	Dither 1	4	4	555.008	
1	SHORT(A)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
2		IMAGER	F1000W	FASTR1	50	1	1	Dither 1	4	4	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	
3		IMAGER	F1130W	FASTR1	50	1	1	Dither 1	4	4	555.008	
3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	
3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	

Special Requirements	
	Aperture PA Range 50 to 100 Degrees (V3 50.0 to 100.0)

Observation

Proposal 3153, Observation 8

Wed May 10 23:18:08 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	2MASS-J06320799-6810419	RA: 06 32 8.0052 (98.0333550d) Dec: -68 10 41.46 (-68.17818d) Equinox: J2000	Proper Motion RA: 0.0016768833489793644 sec of time/yr Proper Motion Dec: 0.033936 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[M dwarfs, Young stellar objects]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FAST	4	1	1	11.1	141558.14

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F770W	FASTR1	45	1	1	Dither 1	4	4	499.507	
1	SHORT(A)	MRSLONG		FASTR1	45	2	1	Dither 1	4	8	1010.115	
1	SHORT(A)	MRSSHORT		FASTR1	45	2	1	Dither 1	4	8	1010.115	
2		IMAGER	F1000W	FASTR1	45	1	1	Dither 1	4	4	499.507	
2	MEDIUM(B)	MRSLONG		FASTR1	45	2	1	Dither 1	4	8	1010.115	
2	MEDIUM(B)	MRSSHORT		FASTR1	45	2	1	Dither 1	4	8	1010.115	
3		IMAGER	F1130W	FASTR1	45	1	1	Dither 1	4	4	499.507	
3	LONG(C)	MRSLONG		FASTR1	45	2	1	Dither 1	4	8	1010.115	
3	LONG(C)	MRSSHORT		FASTR1	45	2	1	Dither 1	4	8	1010.115	

Observation

Proposal 3153, Observation 9

Wed May 10 23:18:08 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(9)	2MASS-J15460752-6258042	RA: 15 46 7.4264 (236.5309433d) Dec: -62 58 5.19 (-62.96811d) Equinox: J2000	Proper Motion RA: -0.006274628706688199 sec of time/yr Proper Motion Dec: -0.061790999939148605 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[M dwarfs, Young stellar objects]
Extended=NO

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1000W	FAST	4	1	1	11.1	141558.18

Template

Primary Channel	Simultaneous Imaging	Imager Subarray
ALL	YES	FULL

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F770W	FASTR1	40	1	1	Dither 1	4	4	444.006	
1	SHORT(A)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
1	SHORT(A)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	
2		IMAGER	F1000W	FASTR1	40	1	1	Dither 1	4	4	444.006	
2	MEDIUM(B)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
2	MEDIUM(B)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	
3		IMAGER	F1130W	FASTR1	40	1	1	Dither 1	4	4	444.006	
3	LONG(C)	MRSLONG		FASTR1	40	1	1	Dither 1	4	4	444.006	
3	LONG(C)	MRSSHORT		FASTR1	40	1	1	Dither 1	4	4	444.006	